17.1	Use sign, Wilcoxon signed-rank or paired t-test, understanding appropriate test selection and interpreting the results in context.	Students will be expected to use the sign and Wilcoxon signed-ranked tests for paired data. Students will be expected to learn and recall necessary conditions for the validity of these tests, and may be required to use these to select the test that is most appropriate in a given context. Sign test – no assumption necessary regarding the distribution for test to be valid. Wilcoxon signed-rank test – assumption that the distribution of differences is symmetrical for test to be valid. Paired t-test – assumption required is that the distribution of differences is normal for test to be valid.

18 Exponential and Poisson distributions

Subject content		Additional information
18.1	Determine when a Poisson model is appropriate (in real world situations including modelling assumptions).	Students will be expected to know the conditions necessary for a Poisson model to be appropriate and to explain or discuss whether these conditions are likely to be present in a given context.
18.2	Determine when an exponential distribution is appropriate (and its relationship to the Poisson distribution as a model of the times between randomly occurring Poisson events).	Students will be expected to know and use the relationship between corresponding Poisson and exponential distributions, in a given context, for Poisson events occurring randomly in time or in space.